

## Zhenqi Zhou, Ph.D. Candidate

Data Scientist | GeoAI and Analytics for Climate, Environment, and Health  
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### EXPERIENCE SUMMARY

Data scientist with 7+ years of experience building end-to-end analytics and AI/ML solutions for climate risk, environmental sustainability and health equity. Skilled in the full lifecycle from data discovery/QA to modeling, validation/monitoring, and decision-ready dashboards, maps and reporting for international stakeholders

### WORK EXPERIENCE

World Bank – IFC, Consultant

05/2024 - Present

#### Integrated Development Analytics – Health, Climate, Environment, and Tourism

- Processed, analyzed, and mapped large-scale datasets (healthcare, climate disasters, tourism, remote sensing, wealth, and population) across low- and middle-income countries (LMICs) using Python, ArcGIS Pro, and Map API to produce evidence-based operational decisions
- [AI] Built predictive models (Linear Regression, Random Forest, XGBoost, Huff Model) for **market demand, climate risk, and service accessibility**, producing outcome metrics adopted by local providers to strengthen resilience and equity in healthcare delivery
- [AI] Developed GPT-based address-matching pipeline to standardize unstructured location and facility data
- Conducted geospatial analysis of 1,000+ healthcare facilities in Kenya and Tanzania, linking facility quality with **climate and economic vulnerabilities** to identify resilience and equity gaps and support **integrated solutions and impact replication**
- Collaborated across IFC and WBG teams to support **outcome measurement** initiatives, developing evaluation dashboards and reproducible data workflows to strengthen **results monitoring**

Weill Cornell Medical College, Cornell University, Research Assistant

08/2024 - 06/2025

#### Real-World Evaluation of Cardiovascular Disease Risk Factors

- Applied fuzzy string matching to geocode patient-level data and aggregate it into neighborhood level for analysis and modeling
- Built scalable Python pipelines for **real-world outcome evaluation** of cardiovascular disease risks across New York City, integrating health and socioeconomic indicators to guide **data-informed policy decisions** [[GitHub repository](#)]

GeoAI Lab, University at Buffalo – SUNY, Research Assistant

09/2021 - 05/2024

#### Climate Disaster Impact Modeling

- Developed data-driven frameworks for assessing spatial and temporal impacts of climate disasters using human mobility data, remote sensing images, and 311 service request data
- [AI] Developed Geographical Random Forests (GRF) in Python which takes spatial index into consideration [[paper](#)]
- [AI] Applied time series models (LSTM, Transformer), spatial econometric models (SLM, SEM, GWR, MGWR), and machine learning models (SVM, RF, GRF) with explainable AI (SHAP) to predict **disaster risks and evaluate differential physical and social vulnerabilities of communities** during climatic events [[GitHub repository](#)]
- Findings **supported local agencies in disaster response** and first-author [paper](#) and [paper](#) published and presented at the 2023 and 2025 AAG Conferences

#### Obesity Prevalence Estimation

- Designed obesity estimation and neighborhood-level activity indicators to measure **community health outcomes**, contributing to **People and Equity outcomes** in urban health analytics
- [AI] Implemented models (OLS, GWR, RF, DNN, GRF) to predict obesity prevalence with accuracy over 90% through cross validation [[GitHub repository](#)]
- First-author [paper](#) published and presented at the 2022 GEOMED Conference

Nanjing Forestry University, Research Assistant

09/2018 - 06/2021

#### Medical Facility Accessibility Assessment

- Led **COVID-19 medical facility accessibility analysis**, processing over 2 billion travel records to design equitable service areas
- Findings informed **data-driven infrastructure and health planning**, aligning with the **People and Infrastructure** knowledge verticals
- First-author [paper](#) published and presented at the 2022 UCGIS Symposium

### TECHNICAL SKILLS

- **Decision support products:** KPI-aligned dashboards, geospatial visualizations and actionable insights
- **Predictive modeling:** Random Forest, XGBoost, spatial ML (Geographical Random Forest), time series (LSTM/Transformer)
- **Reproducible pipelines:** Python (Jupyter), R/SQL, Git, automated QA/QC, API-based data acquisition
- **Specialties:** AI, Cross-Sector Analytics (Health, Climate, Environment), Big Data Integration, Outcome Measurement, Impact Evaluation, GIScience, Remote Sensing

### EDUCATION

University at Buffalo – SUNY, USA

09/2021 - 03/2026 (expected)

Ph.D. in GeoAI and Geospatial Data Science